

DK CAD Pro — USER MANUAL

****How to open panel****

** Type `DKPANEL` or ACTIVATE_DK_CAD in AutoCAD
Select command
Click Execute Tool
Enter key
Activate

1. CREATE TIN MESH MODELER 3D SURFACE

- Layer name for 3D faces [MESH]: *(Enter layer name or press Enter for)*
- Select point entities → **Enter**
- Done. 3D mesh created on specified layer.

2. 3D MESH SURFACE VOLUME CALCULATOR

- 2 different mesh layers must be present in drawing
- Select any one mesh of OGL *(existing ground)*
- Select any one mesh of FRL *(proposed level)*
- Click base point for table
- Cut/Fill volume table created at base point

3. CREATE CROSS SECTIONS

- Mode [Single/Dual Surface]: *(Single = one mesh, Dual = OGL + FGL)*
- OGL layer name *(Enter = window select)*
- FGL layer name *(Enter = window select)*
- Chainage interval (m) e.g. 10
- Left offset limit (m)
- Right offset limit (m)
- Offset step interval (m) e.g. 5
- Datum elevation (m)
- Vertical exaggeration (e.g. 5)
- Camber on profile lines [OGL/FGL/Both/None]
- Camber in bands [OGL/FGL/Both/None]
- Gradient break sensitivity % (e.g. 0.2)
- Pick base point for first cross-section
- Cross sections generated automatically

4. LONGITUDINAL PROFILE BUILDER L-SECTION

- OGL layer name *(Enter = window select)*
- FGL layer name *(Enter = window select)*
- Chainage interval (m) e.g. 10
- Datum elevation (lowest Z, e.g. 20.0)
- Vertical exaggeration (e.g. 10)
- Gradient break sensitivity % (e.g. 0.5)
- Gradient display [FGL/OGL/Both]
- Pick base point for profile
- L-Section profile generated

5. SURFACE ELEVATION HEATMAP

- [1] OGL layer name *(existing ground mesh)*
- [2] FGL layer name *(proposed mesh)*
- [3] Output layer [DIFF_FILL]: *(Enter or type name)*
- Source layers freeze? [Y/N]
- How many ranges? (1-20) *(e.g. 5)*
- For each range: enter value and color number
- Out-of-range color [GREY]: *(Enter or type)*
- Colour heatmap created showing cut/fill difference

6. STATION-OFFSET NODE BUILDER FOR GRID

- Select centerline alignment polyline → **Enter**
- Chainage interval (m): *(e.g. 10)*
- Offset interval (m): *(e.g. 5)*
- Offset mode [B=Boundary / M=Max offset]: *(B or M)*
- Max offset each side (m): *(if M selected)*
- Chainage text height (m):
- Grid nodes created with station-offset labels

Note:- Apply No.6 command for Grid sheet Export.

7. GRID SHEET REPORT CH/OFF/CAMBER/RL

- Select centerline polyline → **Enter**
- CSV file saved to Downloads folder automatically
- Report includes: Chainage, Offset, RL values

Note:- Run No.6 to get Grid sheet Export Report.

8. GRID EARTHWORK ESTIMATOR (VOLUME)

- Select all grid polylines → **Enter**
- Enter X distance of Grid: *(e.g. 5)*
- Enter Y distance of Grid: *(e.g. 5)*
- Enter Text Height:
- Volume calculated per grid cell, total shown

9. DUAL-MESH SURFACE LEVEL VARIANCE

- Choose Mode [Grid / Polyline]:
 - **If Grid:** Enter X spacing, Enter Y spacing
 - **If Polyline:** Select polyline, choose [Vertices / Interval]
 - Enter Text Height:
 - Level difference values shown at each point
- ## ## 10. DYNAMIC SLOPE INTERCEPTOR FRL TO OGL

- Select FRL boundary polyline → **Enter**
- Enter FRL (Elevation) of the Pad: *(e.g. 25.500)*
- Enter Grading Slope (e.g. 2 for 1:2 slope):
- Grading lines created from pad to natural ground

11. CATCHMENT FLOW & SLOPE ANALYZER

- Select 3D mesh → **Enter**
- Specify maximum length of flow arrow: *(e.g. 2)*
- Specify text height for slope labels: *(e.g. 0.5)*
- Slope arrows drawn showing drainage direction

12. SMART CROSSFALL PROFILE OFFSET

- Select profile/centerline polyline → **Enter**
- Enter last chainage or end chainage:
- Enter offset (Width distance): *(e.g. 3.5)*
- Enter Slope (%): *(e.g. 2.5)*
- Select [Left/Right/Both]:
- Enter interval for smooth curves *(only for arcs)*
- Select Insertion Point for Longitudinal Section
- Offset profile with crossfall generated

13. SMART POLY-SYNC 2D TO 3D CONVERTER

- Select tool [2Dto3D / 3Dto2D / ToPL]:
 - **2Dto3D** = Convert 2D polyline to 3D using mesh
 - **3Dto2D** = Flatten 3D polyline to 2D
 - **ToPL** = Convert arcs/lines to polyline
- Enter interval to smooth curves (e.g. 1.0 or 2.0):
- Select objects → **Enter**
- Conversion done

14. CREATE CONTOURS

- Select point objects or mesh → **Enter**
- Enter Contour Interval (e.g. 0.010 or 1.0):
- Enter Text Height:
- Enter distance between texts (e.g. 20 or 50):

- Contours generated with labels

15. NODE POINT INTERPOLATOR AT 3D MESH

- Select 3D mesh → **Enter**
- Select operation mode [Grid/Vertex/Interval/Random]:
 - **Grid** = Enter X spacing, Y spacing, pick origin
 - **Vertex** = Points at every mesh vertex
 - **Interval** = Points at set distance along mesh
 - **Random** = Click on surface to place point
- Points created at mesh elevation

16. PROXIMITY SELECTOR NEAR POLYLINE

- Select reference polyline → **Enter**
- Enter Max Distance (Buffer offset): *(e.g. 5)*
- All objects within that distance selected automatically

17. POLYLINE GEOMETRY ANNOTATOR

- Select polylines → **Enter**
- Each segment annotated with length and bearing
- No further input required

18. POLYLINE GEOMETRY EXPORTER

- Select polylines → **Enter**
- CSV file saved to Downloads automatically
- Includes: Length, Area, Layer, Start/End points

19. CREATE FEATURE LINE

- **NOTE:** Select only 2D polylines with 0 elevation
- Select 2D polylines → **Enter**
- Feature line created on selected site

20. QUICK 3D OBJECT INSPECTOR

- Select any object in drawing
- Properties shown: Type, Layer, Length, Area, Z values
- Report saved to Downloads folder

21. ANNOTATE 3D POLYLINES

- Select 3D polylines → **Enter**
- RL values annotated at each vertex automatically

- No further input required

22. 3D POLYLINE VERTEX RL EXPORTER

- Select 3D polylines → **Enter**
- CSV saved to Downloads automatically
- Includes: Vertex number, X, Y, Z (RL)

23. EXPORT COGO OR SIMPLE POINTS (X,Y,Z)

- Dialog box opens → Select save location
- Select point objects → **Enter**
- CSV created with: Point No, X, Y, Z, Description

24. ANNOTATE MULTIPLE POINT X Y Z

- Select point objects → **Enter**
- X, Y, Z values annotated at each point automatically

25. ANNOTATE Z VALUE

- Enter Text Height <2.0>: *(Enter or type height)*
- Click points one by one in drawing
- Z (RL) value annotated at each clicked location
- Press **Enter** to stop

26. ANNOTATE SINGLE POINT X Y Z

- Pick coordinate point: *(click in drawing)*
- X, Y, Z values annotated at that point
- One click = one annotation

27. ANNOTATE BLOCK

- Select block objects → **Enter**
- Block name annotated at each block automatically

28. BLOCK ATTRIBUTE EXTRACTOR

- Select block objects → **Enter**
- CSV saved to Downloads automatically
- Includes: Block name, Insert point, Attributes

29. CONVERT COGO TO BASIC NODE POINTS

- Select Civil 3D COGO points → **Enter**
- Basic AutoCAD points created at same locations
- Works in all AutoCAD versions

30. POINT TO 3D POLYLINE

- Select point objects in order → **Enter**
- 3D polyline drawn through all selected points
- Elevation taken from point Z value

31. IMPORT TEXT TO CAD (FROM CSV)

- Prepare CSV file: Columns = X, Y, Z, Text
- Command runs → Select CSV file
- Text inserted at each coordinate in drawing

32. IMPORT TEXT WITH MLEADER FROM CSV

- Prepare CSV file: Columns = X, Y, Z, Text
- Command runs → Select CSV file
- MLeader annotations inserted at each point

33. EXPORT TEXT TO CSV

- Dialog box opens → Select save location
- Select text/mtext objects → **Enter**
- CSV created with: X, Y, Z, Text content

34. TEXT TO NODE CONVERTER

- Select all TEXT/MTEXT objects → *(auto selected)*
- Points created at text insertion locations
- Z taken from text numeric value if applicable

35. INSERT TEXT VALUES TO TEXT Z

- Select TEXT/MTEXT objects → *(auto selected)*
- Z elevation of each text set to its numeric value
- Useful for elevating spot level texts to their RL

36. CREATE & EXPORT ARL DRL DIFF REPORT

- Select ARL point objects *(Actual RL)* → **Enter**
- Select DRL reference *(Design RL)* → **Enter**
- Difference calculated at each point
- Report CSV saved to Downloads automatically

37. CREATE POINTS AT INTERSECTION (LINES)

- Select all lines/polylines → **Enter**
- Points created at every intersection automatically
- No further input required

38. CREATE POINTS AT INTERSECTION (FIX)

- Select intersecting lines/polylines → **Enter**
- Lines precisely trimmed/extended to intersection
- Points created at intersection locations

QUICK TIPS

- CSV files are always saved to **Downloads** folder
- Press **Enter** after selecting objects
- **Escape** to cancel any command
- All commands work in AutoCAD 2020 to 2027

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